

DTSA 5714

Applications of Software Architecture for Big Data

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (89.6 / 100)	74.9 / 100
SPECIALIZATION Fundamentals of Software Architecture for Big Data		

75

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+37

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Intended for individuals who want to build a production-quality software system that leverages big data. Students will apply the basics of software engineering and architecture to create a production-ready distributed system that handles big data. Students will build and scale a large, data intensive, distributed system, composed of loosely coupled, highly cohesive data microservices.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	89.6	100.0	+10.4	LOW
MLOps Engineer	78.6	100.0	+21.4	MEDIUM
ML Engineer	77.7	100.0	+22.3	MEDIUM
Machine Learning Engineer	76.9	100.0	+23.1	MEDIUM
Applied Scientist	63.9	100.0	+36.1	HIGH
Data Scientist	62.9	100.0	+37.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Model Deployment & Serving 2 of 6 roles

production

LLMs & Generative AI 1 of 6 roles

storage

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.
- LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

GAP KEYWORDS BY ROLE

AI Engineer

89.6 → 100.0 (+10.4 pts)

learning infrastructure technical code systems teams based problems

MLOps Engineer

78.6 → 100.0 (+21.4 pts)

infrastructure production text systems learning machine tools error

ML Engineer

77.7 → 100.0 (+22.3 pts)

discovery learning machine systems production starting infrastructure world

Machine Learning Engineer

76.9 → 100.0 (+23.1 pts)

sets learning analytics tools solutions business mobile time

Applied Scientist

63.9 → 100.0 (+36.1 pts)

lab discovery scientific computational research storage problems technical

Data Scientist

62.9 → 100.0 (+37.1 pts)

rules sets characteristics analytics evolution scripting insights analytical

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.